Best Ice Cream Flavours (`icecream`)

Filippo loves ice cream, so when he found out that a new ice cream shop had opened in Bologna, he had to go and taste their ice cream!

Figure 1: Choosing the right flavour is always difficult.

The shop sells N flavours of ice cream, each flavour is represented by a string F_i. The ice cream shop offers a special promotion, each customer can choose a non-empty string s and he will receive a cup containing **all flavours** starting with s, for example if the flavours are "chocolate", "coffee" and "cookies", by choosing the string "co" you would receive the flavours "coffee" and "cookies" but not the "chocolate" flavour.

Filippo wants to buy exactly k flavours using the fewest number of cups, however Filippo can't decide which flavours to choose so, to avoid blocking the queue, he decided to buy exactly the first k flavours (i.e. flavours $F_0, F_1, \ldots, F_{k-1}$).

Help Filippo find the minimum number of cups needed to buy exactly the first k flavours, for each $k = 1, \ldots, N$.

> ☞ Among the attachments of this task you may find a template file `icecream.*` with a sample incomplete implementation.

Input

The first line contains the only integer N. The next N lines contain a string representing a flavour.

Output

You need to write N lines with an integer each: the minimum number of cups needed to order all flavours from 0 to $k-1$ but none of the flavours from k to $N-1$, for each $k = 1, \ldots, N$.

Constraints

- $2 \leq N \leq 100\,000$.
- Each flavour is formed by lowercase English characters (‘a’ to ‘z’).
- The total length of all flavours is at most $2\,000\,000$.
- All flavours are different.
- It’s guaranteed that it is always possible to buy the first k flavours for each $k = 1, \ldots, N$.
- The same flavour is allowed to appear in more than one cup (while still counting as a single flavour).

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

– **Subtask 1** (0 points) Examples.

– **Subtask 2** (22 points) $N \leq 10$, the total length of all flavours is at most 100.

– **Subtask 3** (24 points) $N \leq 500$, the total length of all flavours is at most $10\,000$.

– **Subtask 4** (23 points) The flavours are sorted in lexicographical order.

– **Subtask 5** (31 points) No additional limitations.

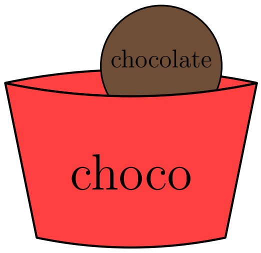
Examples

input	output
5 chocolate coffee cookies pistachio strawberry	1 2 1 2 3
10 orange peanutbutter mango mint cheesecacke coconut oreo cherry peppermint watermelon	1 2 3 3 4 5 5 4 4 5

Explanation

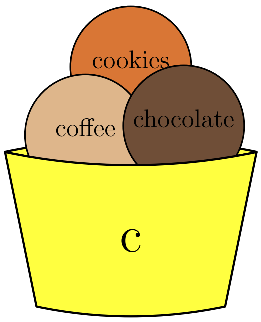
In the **first sample case**:

- $k = 1$: Filippo can buy one cup with s_1 =“choco”.

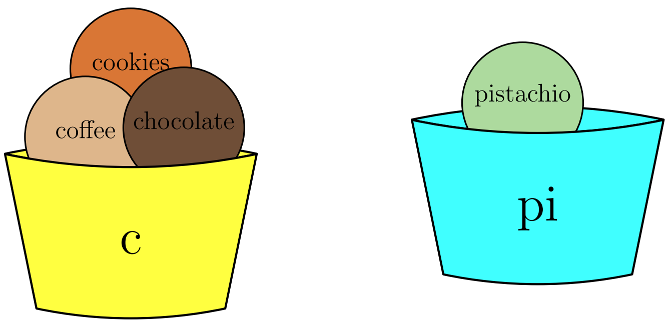


- $k = 2$: Filippo can buy two cups with s_1 =“choco” and s_2 =“coffee”.

- $k = 3$: Filippo can buy one cup with s_1 =“c”.



- $k = 4$: Filippo can buy two cups with s_1 =“c” and s_2 =“pi”.



- $k = 5$: Filippo can buy three cups with s_1 =“c”, s_2 =“pi” and s_3 =“straw”.